Python Programming: Lab Activity 2

Due Date: TBD

Objective

This lab focuses on basic Python programming concepts such as arithmetic calculations, user input handling, and conditional logic. The exercises aim to strengthen your ability to apply formulas, work with user inputs, and implement decision-making logic.

Assignment Tasks

1. Calculate Distance Traveled (3 points)

Write a program to calculate and display the distance a car travels using the formula:
Distance = Speed × Time

Requirements:

- Assume the car is traveling at 70 miles per hour.
- Display the distance the car will travel for:
  - **6 hours**
  - **10 hours**

2. Calculate Miles-Per-Gallon (MPG) (3 points)

Write a program to calculate a car's MPG using the formula:
MPG = Miles Driven / Gallons of Gas Used

Requirements:

- Ask the user to input:
  - Number of miles driven.
  - Gallons of gas used.
- Calculate and display the car's MPG.

3. Convert Seconds to Minutes and Seconds (5 points)

Write a program that converts a user-provided number of seconds into minutes and seconds.

Requirements:

- If the entered seconds are **≥ 60**, convert them into minutes and remaining seconds.
- Display the results in the format: `X minutes, Y seconds`.

4. Book Club Points System (5 points)

Write a program to calculate the points awarded by a book club based on the number of books purchased in a month.

Points Awarding System:

- **0 books**: 0 points
- **1-2 books**: 5 points
- **3-4 books**: 15 points
- **5-6 books**: 30 points
- **7 or more books**: 60 points

Requirements:

- Ask the user to input the number of books purchased.
- Display the corresponding points awarded.

5. Day of the Week (5 points)

Write a program that maps a user-provided number (1 through 7) to the corresponding day of the week.

Mapping:

- **1 = Monday**
- **2 = Tuesday**
- **3 = Wednesday**
- **4 = Thursday**
- **5 = Friday**
- **6 = Saturday**
- **7 = Sunday**

Requirements:

- Prompt the user for a number between **1 and 7**.
- Display the corresponding day.
- Display an **error message** and return None if the number is outside the range.

Submission Guidelines

1. Use the template file provided: **lab2.py**.
2. Write your code in the designated sections of the template.
3. Do **not** rename the template file or modify its structure outside the designated areas.
4. Submit your completed file as instructed.

Grading Criteria

- **Correctness:** Each task is implemented as per the instructions.
- **Output Accuracy:** Results must exactly match the specified requirements.
- **Code Quality:** Solutions should be clear, concise, and well-documented.
- **Originality:** Ensure your work is unique—plagiarism will lead to penalties.

- **AutoGrader Compliance:** Outputs must adhere to the specified format for full marks.